A CASE-STUDY BASED REPORT

ON

FOOTBALL MATCH RESULT PREDICTION USING MULTINOMIAL LOGISTIC REGRESSION

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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Abstract

Football match outcomes can be analysed using historical match data and machine-learning classification. The model uses the input columns available in the football match dataset, such as pH, hardness, dissolved solids, chloramines, sulphate, conductivity, organic carbon, trihalomethanes, and turbidity. Machine Learning provides a data-driven approach for analysing these parameters and automatically classifying football matches according to their match result. The project Football Match Result Prediction Using Multinomial Multinomial Logistic Regression is developed using Python and the Football Match Dataset. The dataset contains 3,276 football matches and 9 numerical football-match parameters, with Match Result as the target variable. The target consists of two classes, where 0 represents non-home win water and 1 represents home win water. The project performs dataset inspection, missing-value analysis, duplicate checking, exploratory data analysis, missing-value imputation, train-test splitting, feature scaling, model training, prediction, and performance evaluation. Two supervised machine-learning algorithms, Multinomial Logistic Regression and Multinomial Multinomial Logistic Regression, are implemented for binary classification. Multinomial Logistic Regression is used as a simple statistical classification model, while Multinomial Multinomial Logistic Regression provides an interpretable rule-based classification approach. The classifier can be evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. The implementation also generates model-comparison graphs, confusion matrices, Multinomial Multinomial Logistic Regression visualization, and feature-importance analysis. Based on the current execution of the project, the Multinomial Multinomial Logistic Regression provides better overall classification performance than Multinomial Logistic Regression, particularly in identifying home win samples. The project demonstrates how Machine Learning can be applied to football-match data to support automated water-match result classification.

Keywords: Football Match Prediction, Machine Learning, Multinomial Logistic Regression, Classification, Exploratory Data Analysis, Match Result

1. Introduction

1.1 Background

Water quality is important for determining whether water is suitable for consumption. Machine Learning can analyse different football-match parameters and identify patterns to classify water as home win or non-home win.

1.2 Motivation for the Project

The motivation is to apply Machine Learning techniques to football-match classification and compare Multinomial Logistic Regression and Multinomial Multinomial Logistic Regression models using the Football Match Dataset.

1.3 Overview of the Work Carried Out

The project involves:

- Loading the Water match result dataset
- Data preprocessing
- Exploratory Data Analysis
- Training Multinomial Multinomial Logistic Regression and Multinomial Logistic Regression
- Analysing feature importance
- Predicting football match result

1.4 Scope and Responsible-Use Statement

The project is limited to classification using the Football Match Dataset and the selected Machine Learning models. It is intended for academic purposes and should not replace professional laboratory football-match testing.

2. Problem Statement

2.1 Case-Study Scenario

A dataset containing different football-match measurements is used to determine whether a football match is home win or non-home win.

2.2 Central Case-Study Question

Can Multinomial Logistic Regression and Multinomial Multinomial Logistic Regression models accurately classify football matches as home win or non-home win using the given football-match parameters?

2.3 Formal Problem Definition

Given 9 numerical football match features, predict the football match result

- 0 → Non home win
- 1 → home win

2.4 Objectives that Guide the Study

- Prepare the Football Match Dataset.
- Perform exploratory data analysis (EDA).
- Prepare the data and handle missing values where required.
- Build a Multinomial Logistic Regression model.
- Build a Multinomial Multinomial Logistic Regression model.
- Evaluate predictions using suitable classification metrics.
- Identify the model's strengths and limitations from its evaluation.

2.5 Assumptions and Limitations

Assumptions:

- The dataset is representative of the football-match classification problem.
- The selected football-match parameters contain sufficient information for classification.
- The training and testing split provides a reasonable evaluation

Limitations:

- The project uses only the Football Match Dataset.
- Only Multinomial Logistic Regression and Multinomial Multinomial Logistic Regression are evaluated.
- The model should not be used as a replacement for professional football-match testing.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

- **Dataset: Football Match Dataset**
- **Samples: Depends on the supplied dataset**
- **Input Features: The selected columns in the supplied dataset**
- **Target: Match Result**
- **Classes: Potable and non-home win**

3.2 Purpose of EDA

EDA is performed to understand the dataset, identify data-quality issues, examine feature distributions, study class distribution, and understand relationships between football match parameter features.

3.3 Preparation Steps Actually Performed

- Loaded the Football Match Dataset.
- Inspected rows, columns and data types.
- Checked for missing values and duplicate records.
- Removed duplicate records if present.
- Separated the input features and match-result target variable.
- Handled missing values using median imputation.
- Divided the data into 80% training and 20% testing sets.

3.4 EDA Exploration – Implementation

- Statistical summary of football-match parameters.
- Match-result class distribution analysis.
- Histograms for selected feature distributions.
- Box plots to inspect possible outliers in numeric features.
- Correlation heatmap to inspect relationships among numeric features.

3.5 Summary of EDA Findings

EDA provided an understanding of the football-match parameters, missing values and class distribution. The analysis also helped identify relationships between the different features.

4. Algorithm Selection and Implementation

4.1 Multinomial Multinomial Logistic Regression

Multinomial Multinomial Logistic Regression classifies football matches using a sequence of feature-based decision rules and provides an interpretable classification approach.

4.2 Multinomial Logistic Regression

Multinomial Logistic Regression is used as a binary classification model to predict whether water is home win or non-home win.

4.3 Model Training and Prediction

The dataset is divided into training and testing subsets. The model is fitted on the training subset and evaluated on held-out test data.

4.4 Performance Evaluation

The model is evaluated using:

- Accuracy
- Recall
- Precision

- Confusion Matrix
- F1-Score

4.5 Feature Importance

Feature importance is analysed using the Multinomial Multinomial Logistic Regression to identify the football-match parameters that contribute to classification.

5. Conclusion

This project successfully developed a machine-learning system for classifying water as home win or non-home win. Multinomial Logistic Regression and Multinomial Multinomial Logistic Regression models were implemented and evaluated using multiple performance metrics. The results show that the Multinomial Multinomial Logistic Regression performed better than Multinomial Logistic Regression for the current classification task. The project demonstrates the application of Machine Learning and EDA techniques to football-match classification.